Table 12.2 Relationships between documents and the envelope shape *(continued)*

Value of `relation`	Description
`contains`	Returns documents if they contain the geometrical shape
`disjoint`	Returns documents that do not exist in the given geometrical shape

In figure 12.6, if we specifically set `relation=intersects`, the expected plots are A, B, and C (as C intersects the main envelope). If we set `relation=within`, plots A and B are returned because they are within the bounded envelope. Setting `relation=contains` returns plots A and B because they are contained in the envelope; and it should not be a surprise that plot D results from `relation=disjoint`.

So far, we've looked at geospatial queries: queries on geodata represented by `geo_point` and `geo_shape` fields. These queries enable us to search for data on a map for various use cases. Now, let's shift gears and discuss queries to search two-dimensional (2D) shapes. In the next section, we explore `shape` queries in detail.

12.7 The shape query

We build 2D shapes such as lines, points, and polygons using x and y Cartesian coordinates. Elasticsearch provides indexing and searching of 2D objects using `shape` queries. For example, a civil engineer's blueprint data, a machine operator's CAD (computer-aided designs) designs, and others fit this criterion. In this section, we briefly index and search geoshapes.

When we work with 2D data, we use the dedicated `shape` data type. We create fields of this `shape` type for indexing and searching 2D data. This is easy to understand by going over the following examples.

The following query shows the mapping for the `myshapes` index with two properties, `name` and `myshape`. As you may notice, the `myshape` attribute is defined as the `shape` data type.

Listing 12.11 Index mapping with a `shape` type

```
PUT myshapes
{
  "mappings": {
    "properties": {
      "name":{
        "type": "text"
      },
      "myshape":{
        "type": "shape"
      }
    }
  }
}
```

Now that we have the mapping, the next step is to index documents with different shapes. The following query indexes two documents with point and line shapes.